Banking AI Assistant

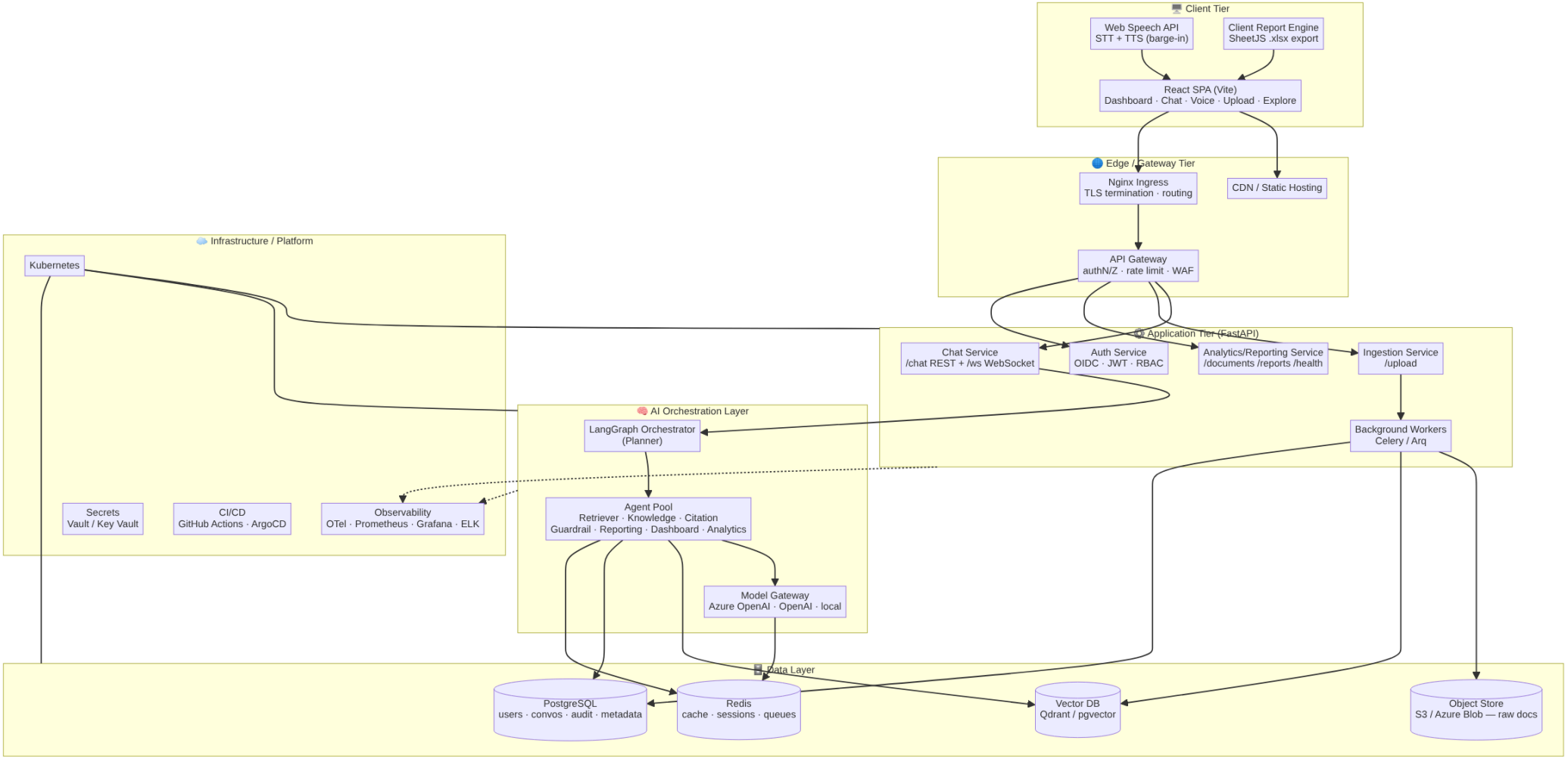
Architecture Diagrams

Enterprise Technical Design — Reference Flowcharts

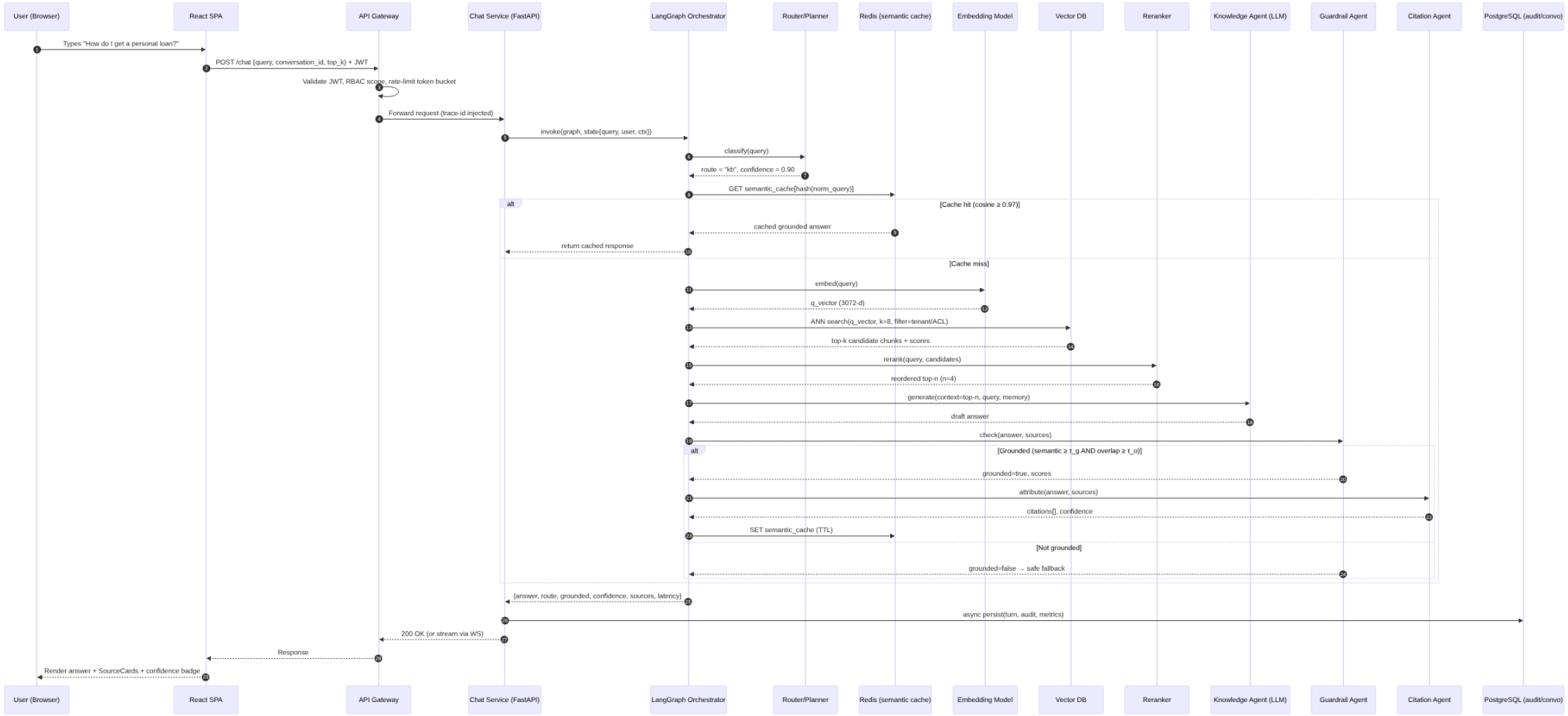
21 diagrams

RAG · Query Routing · Hallucination Guard · Agentic Orchestration

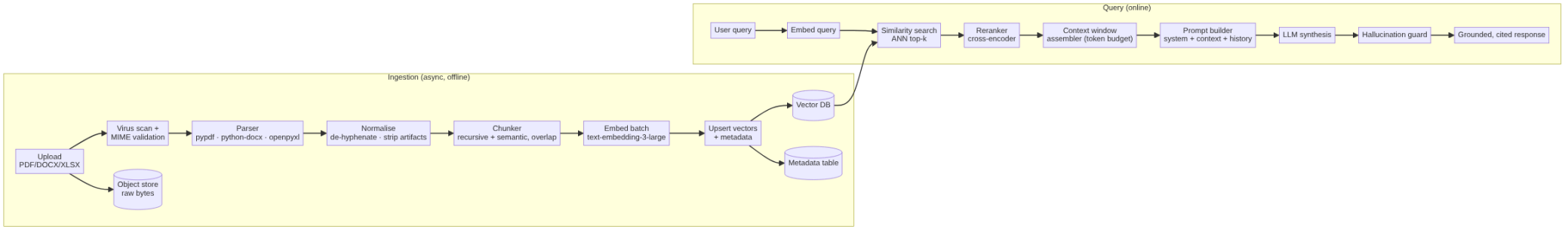
1. High-Level Architecture



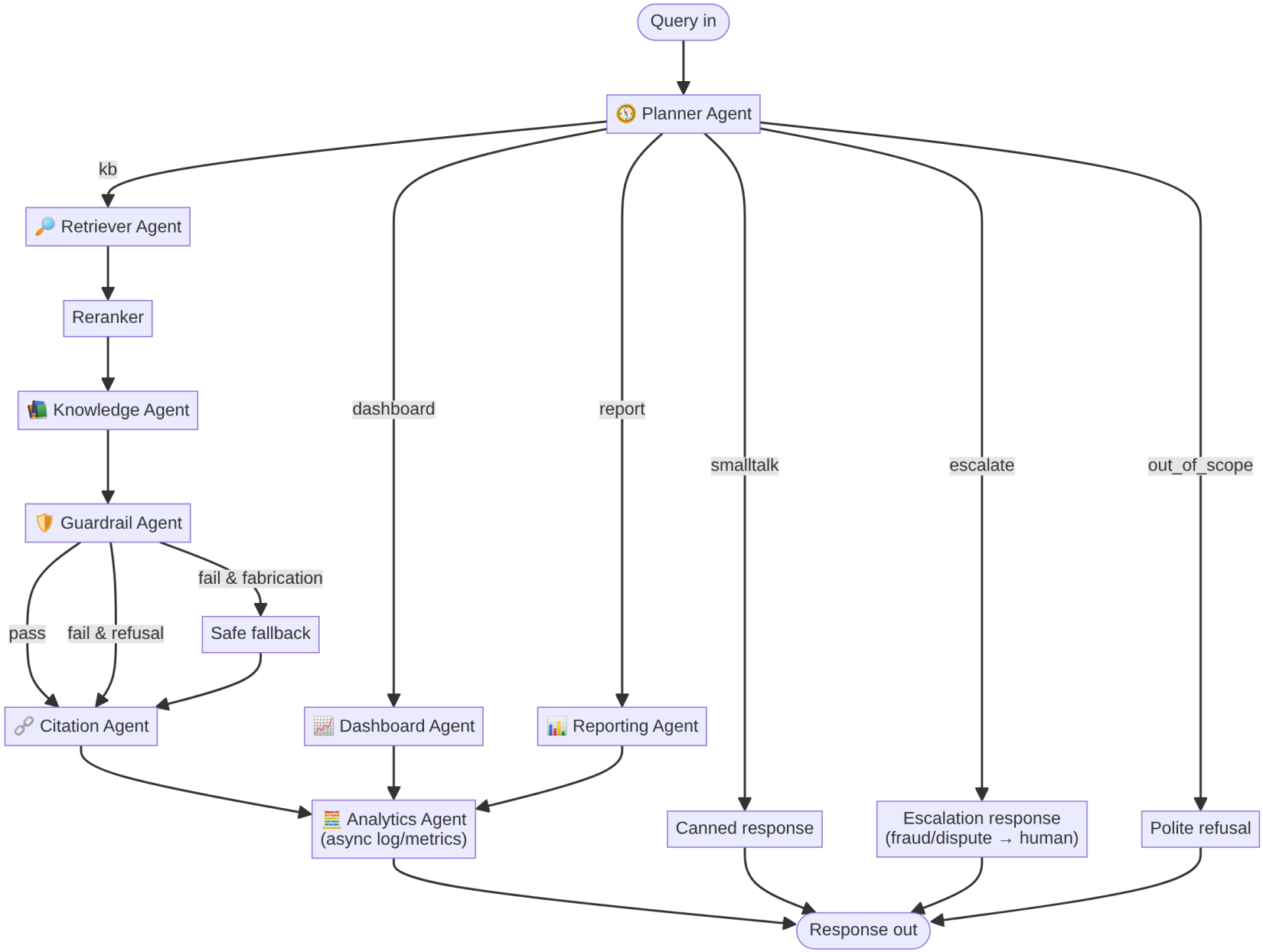
2. End-to-End Request Flow — *"How do I get a personal loan?"*



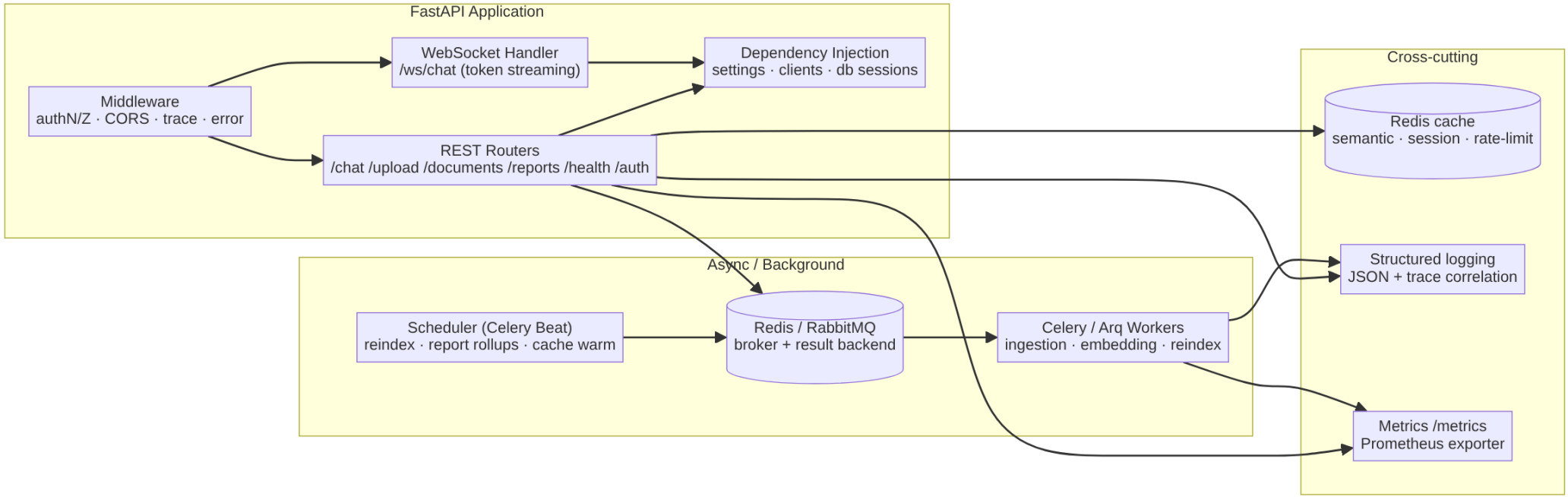
3. RAG Pipeline



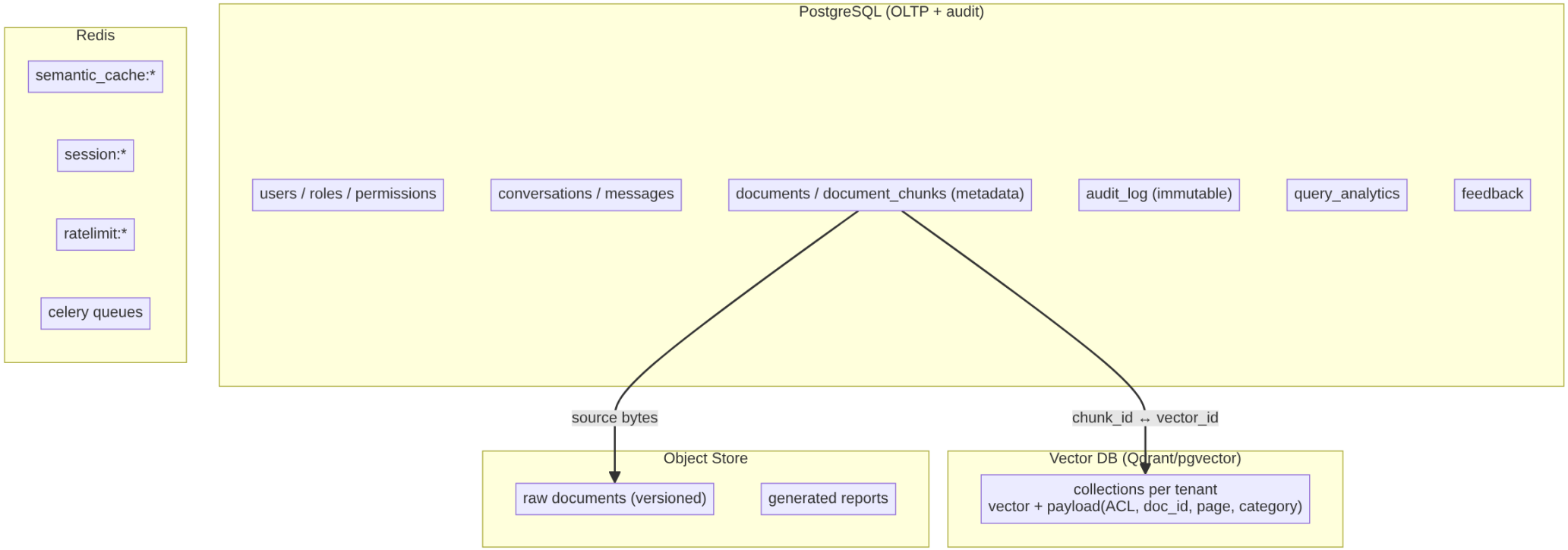
4. Agent Architecture



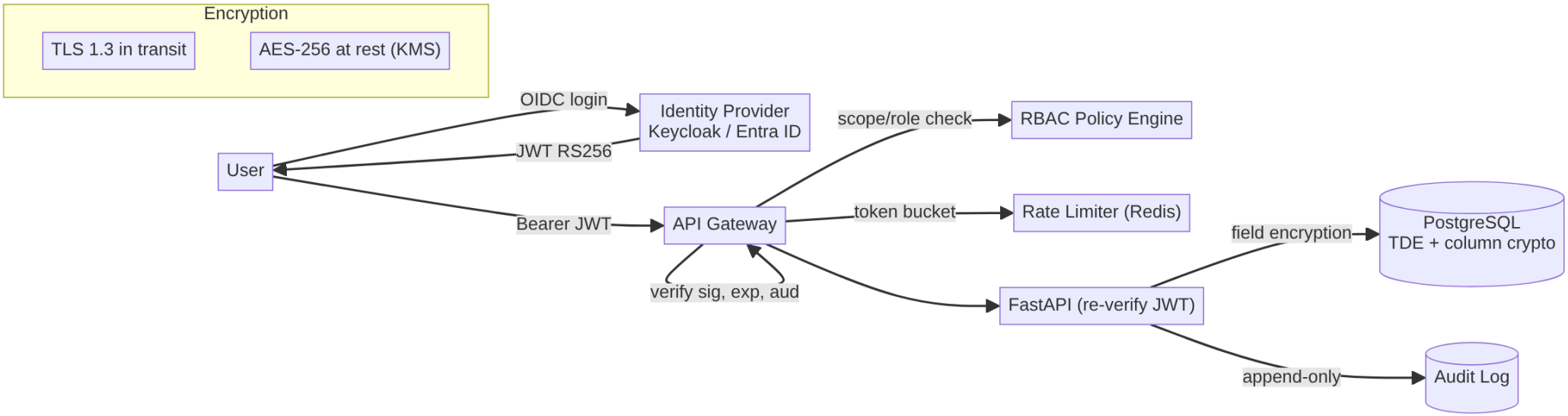
5. Backend Components



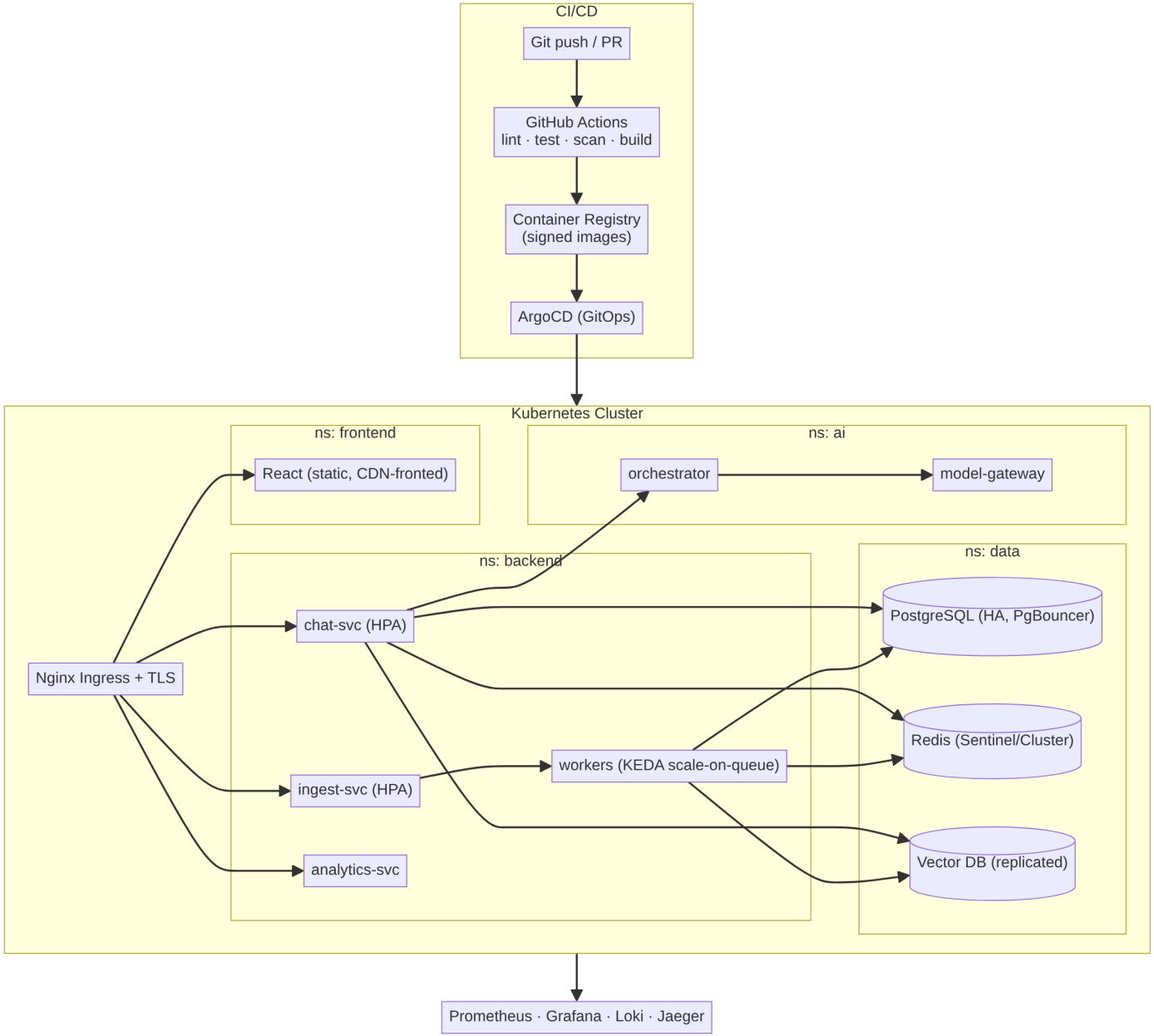
6. Database Design



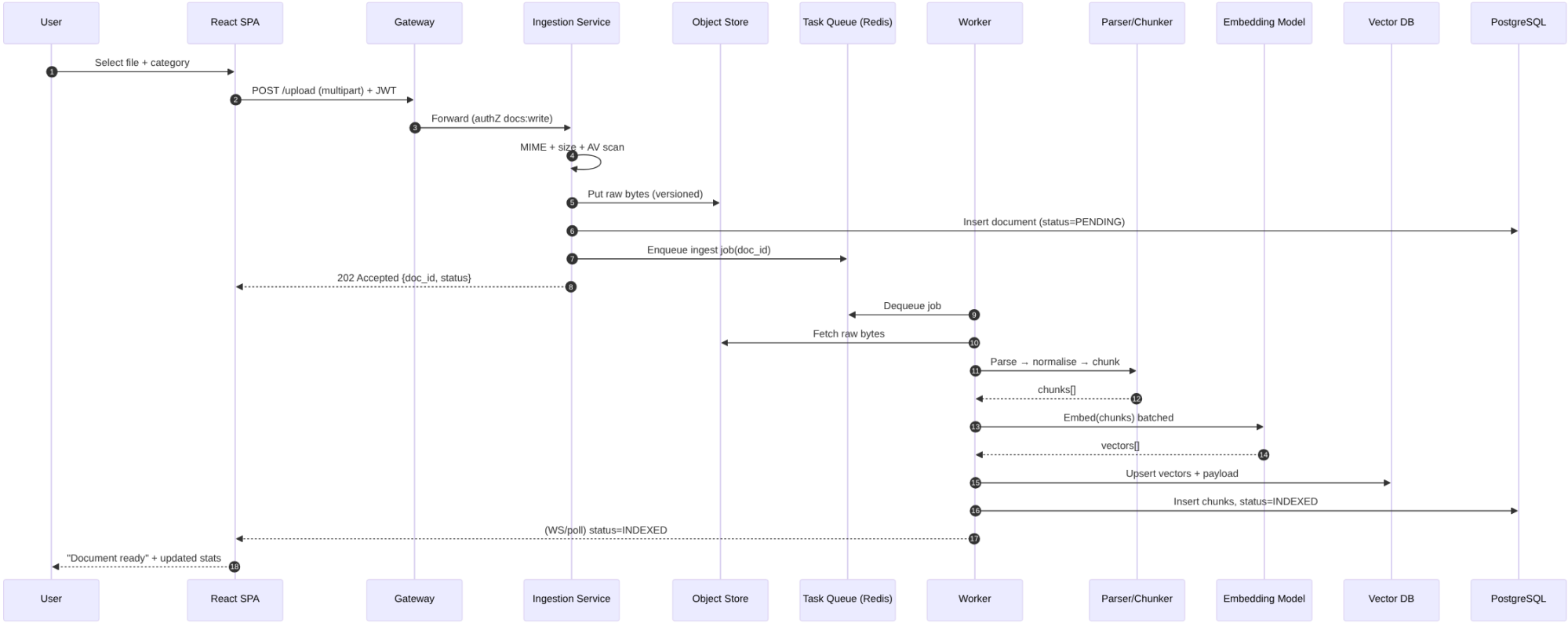
8. Security


Encryption
TLS 1.3 in transit
AES-256 at rest (KMS)
OIDC login
Identity Provider
Keycloak / Entra ID
JWT RS256
User
Bearer JWT
API Gateway
scope/role check
RBAC Policy Engine
token bucket
Rate Limiter (Redis)
verify sig, exp, aud
field encryption
PostgreSQL
TDE + column crypto
FastAPI (re-verify JWT)
append-only
Audit Log

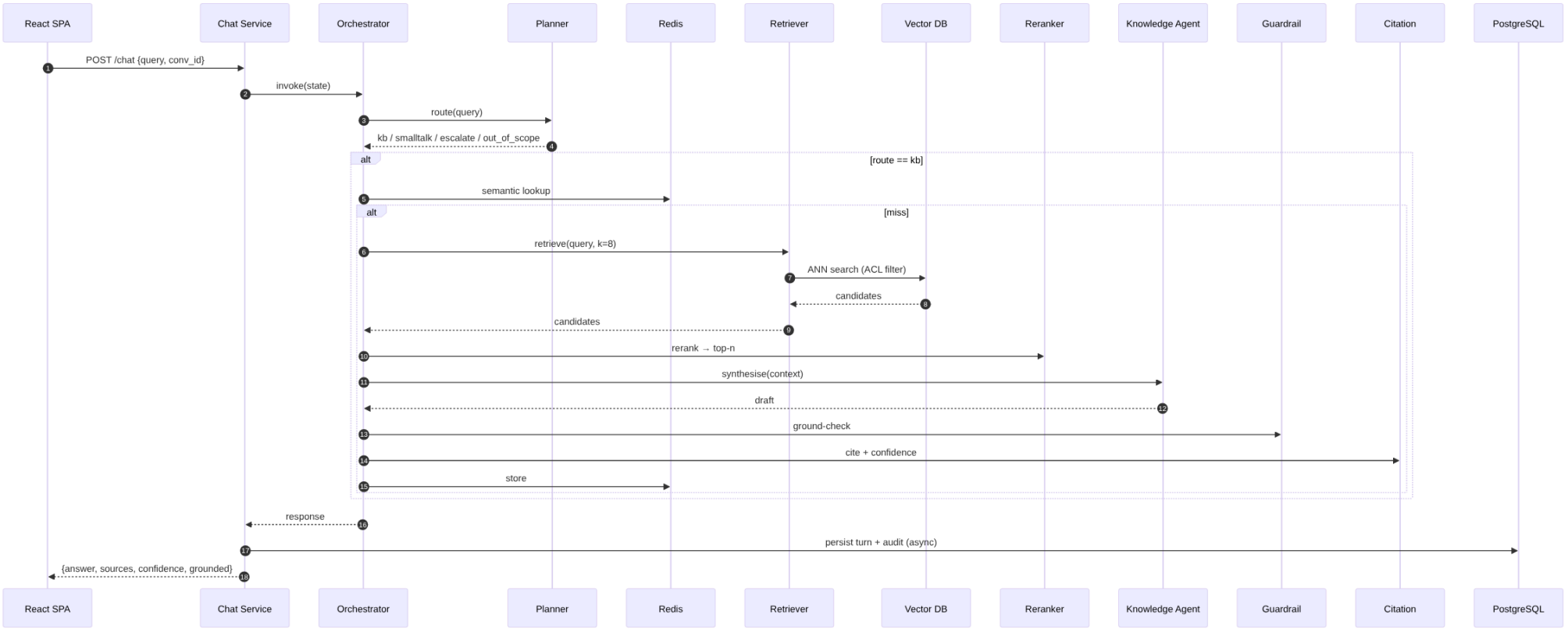
9. Deployment Architecture



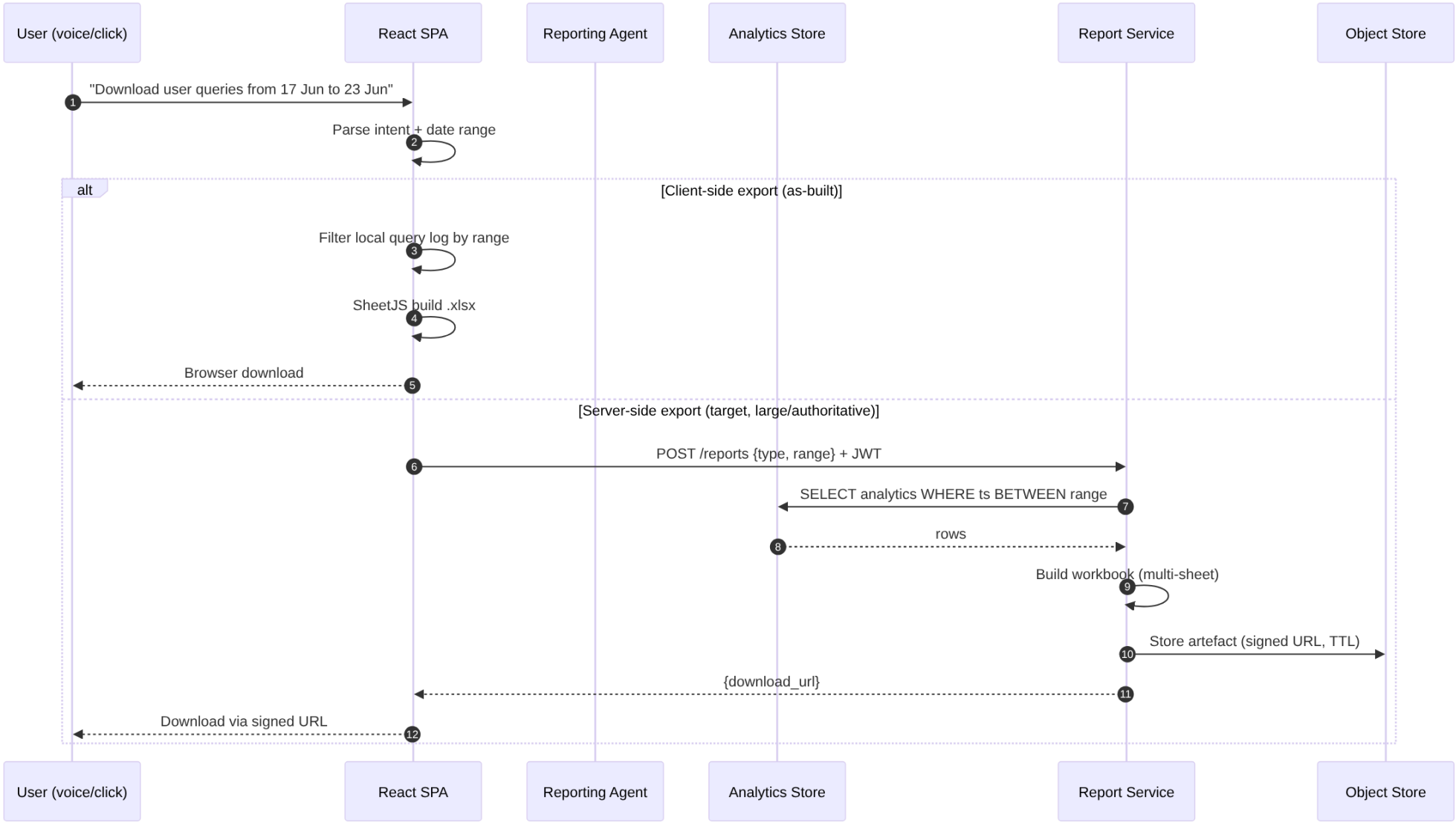
10.1 Document Upload



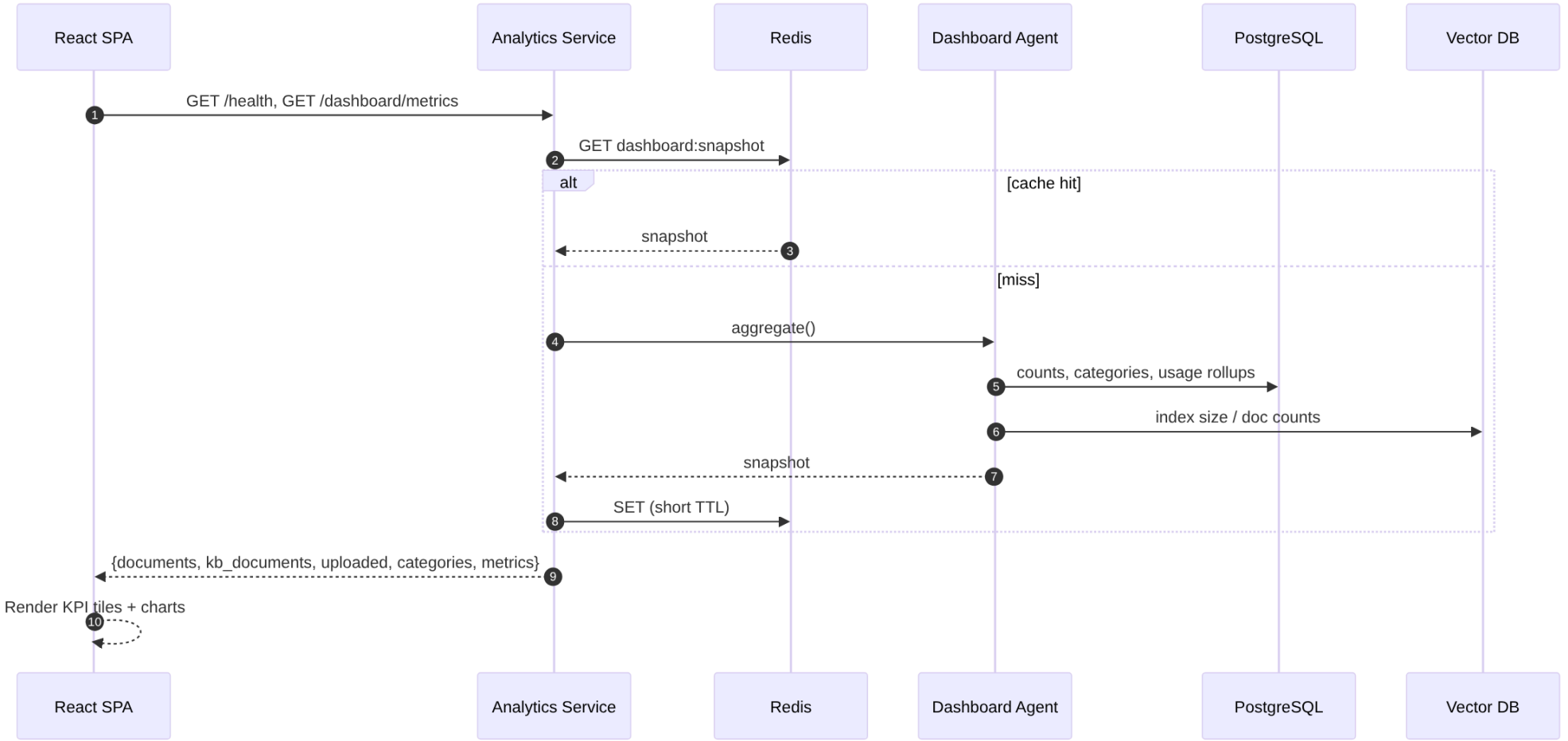
10.2 Question Answering



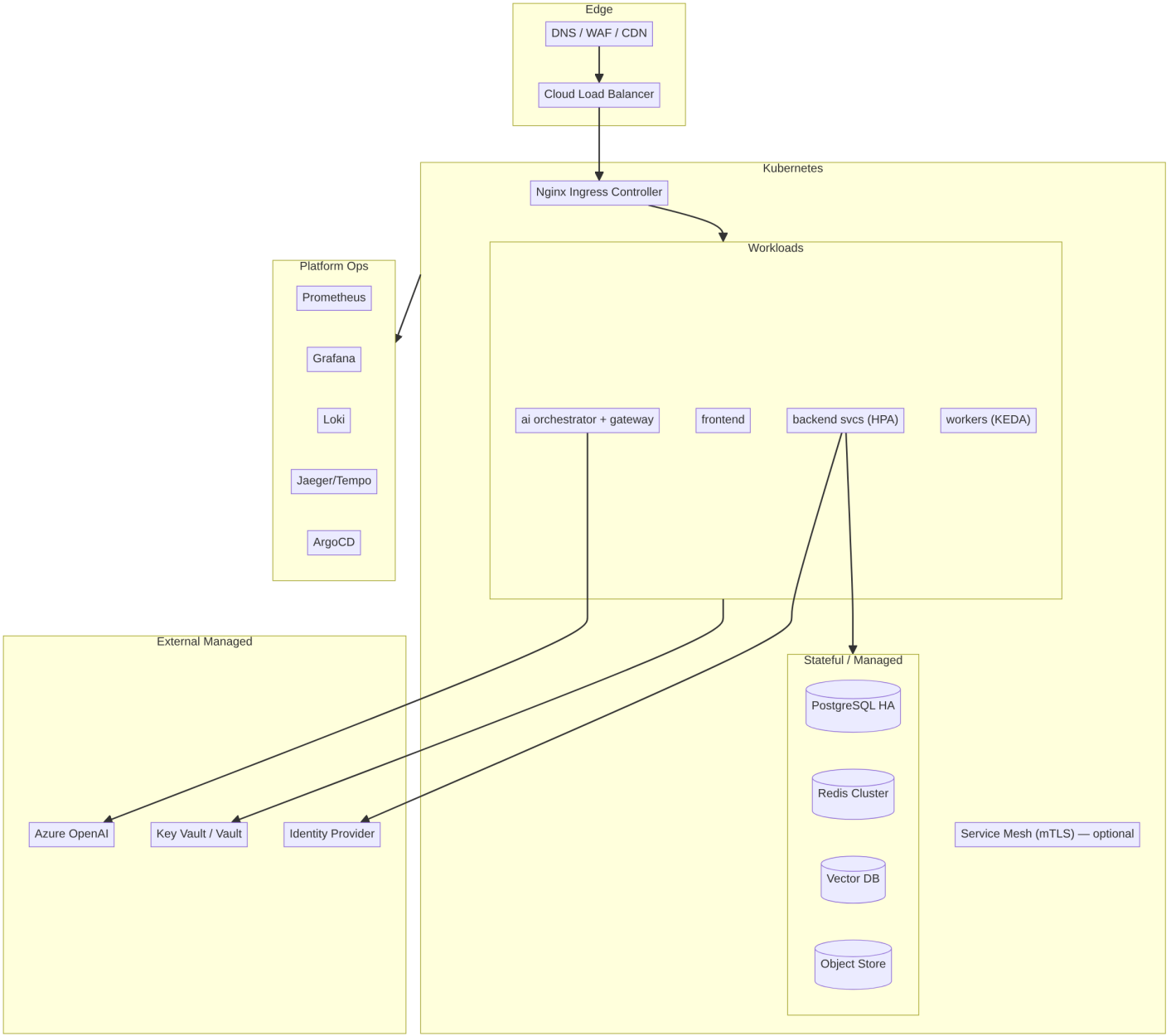
10.3 Report Generation



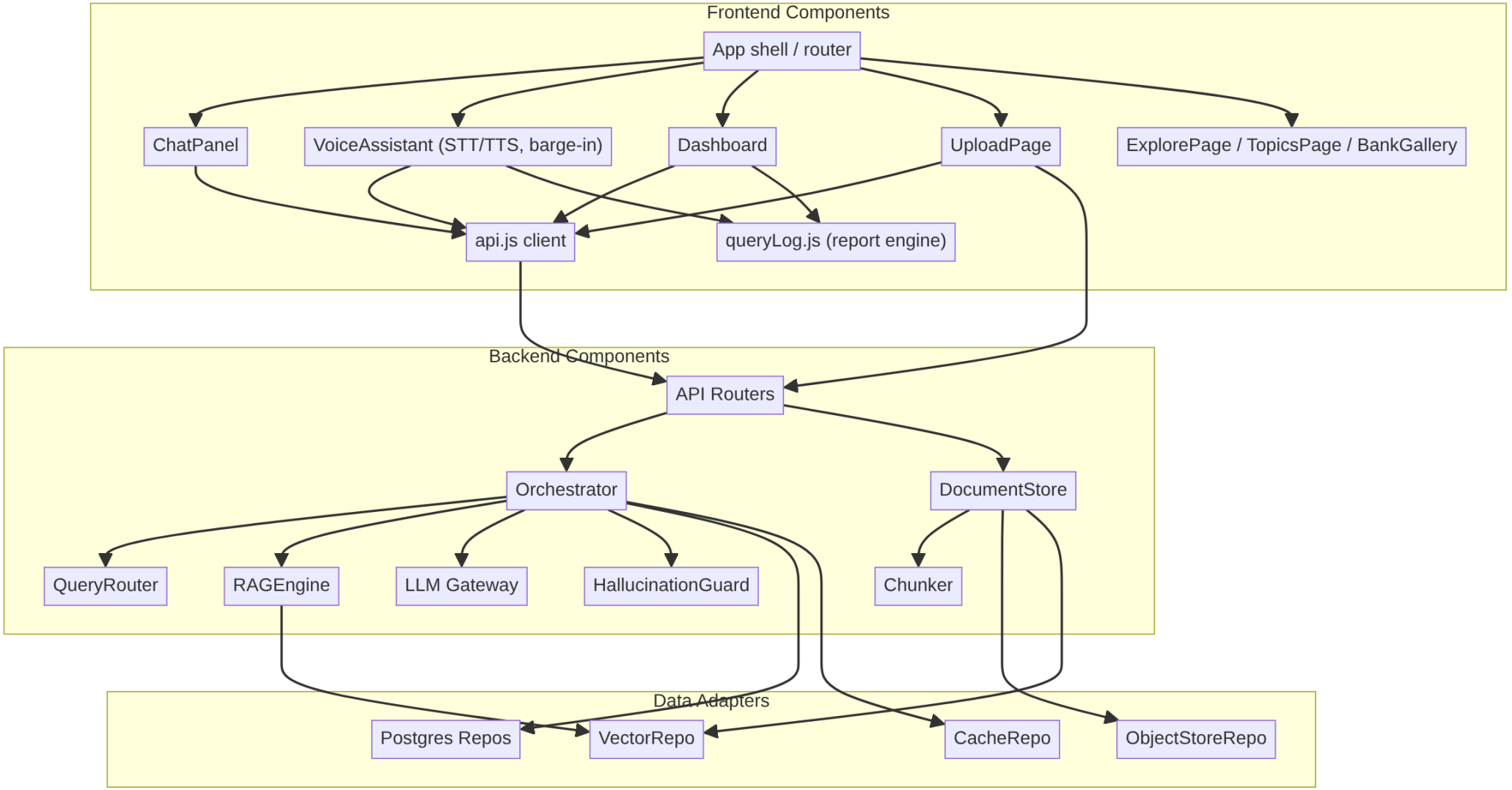
10.4 Dashboard Query



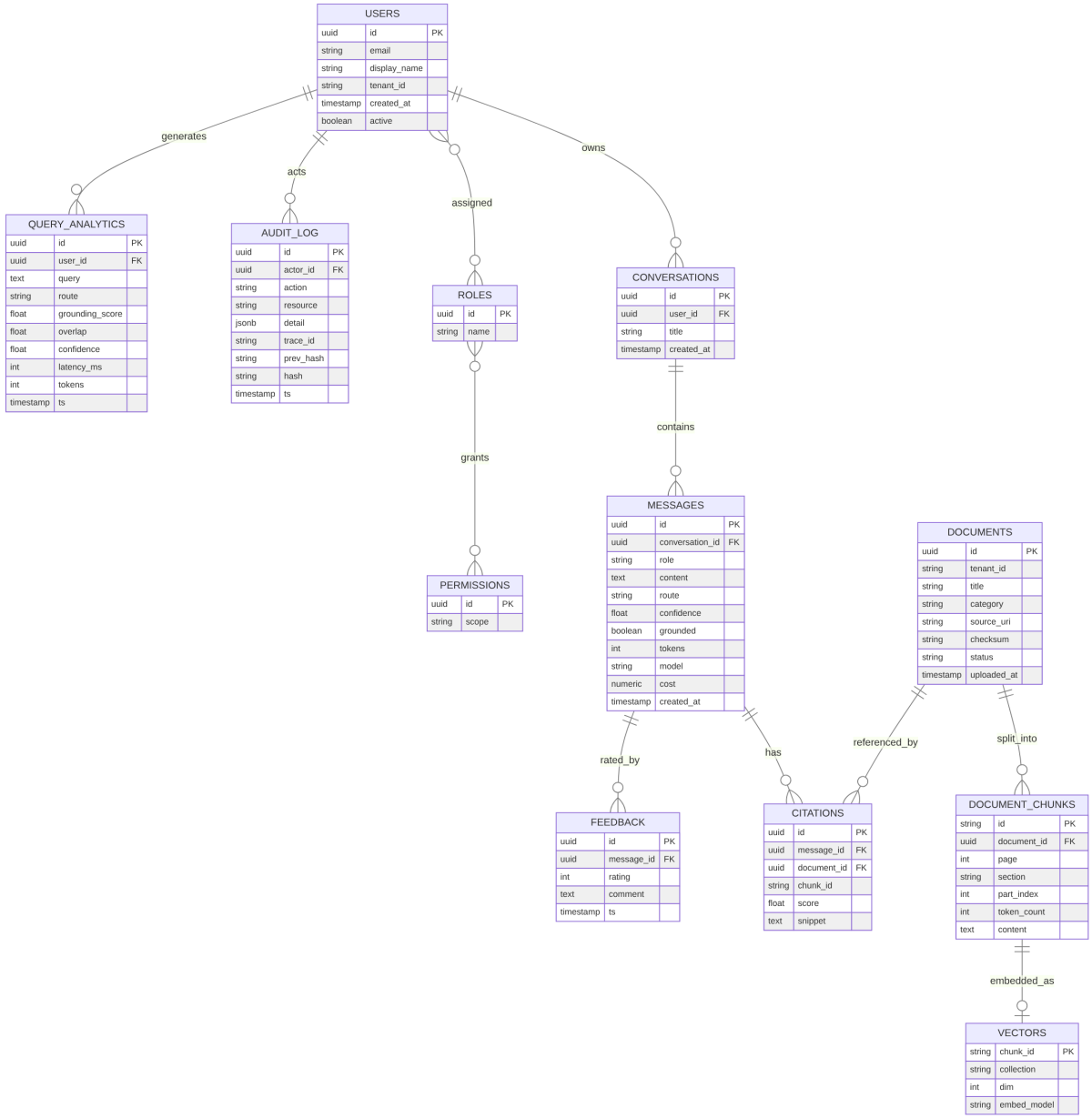
11. Infrastructure Diagram



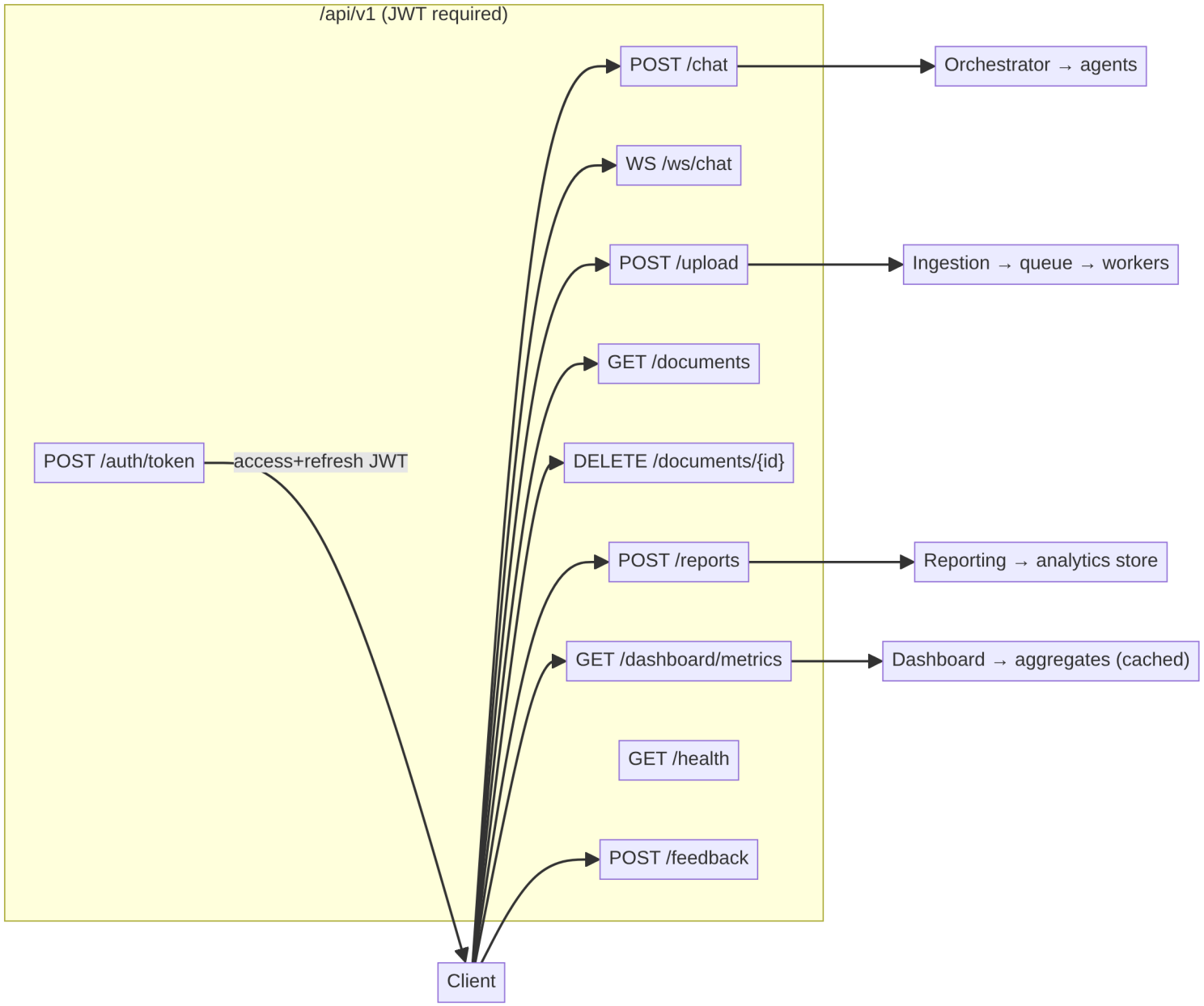
12. Component Diagram



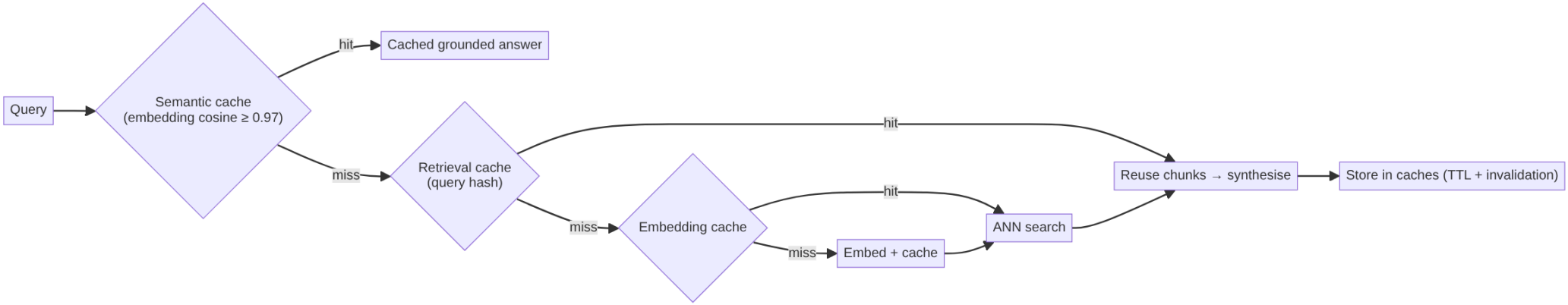
13. Database ER Diagram



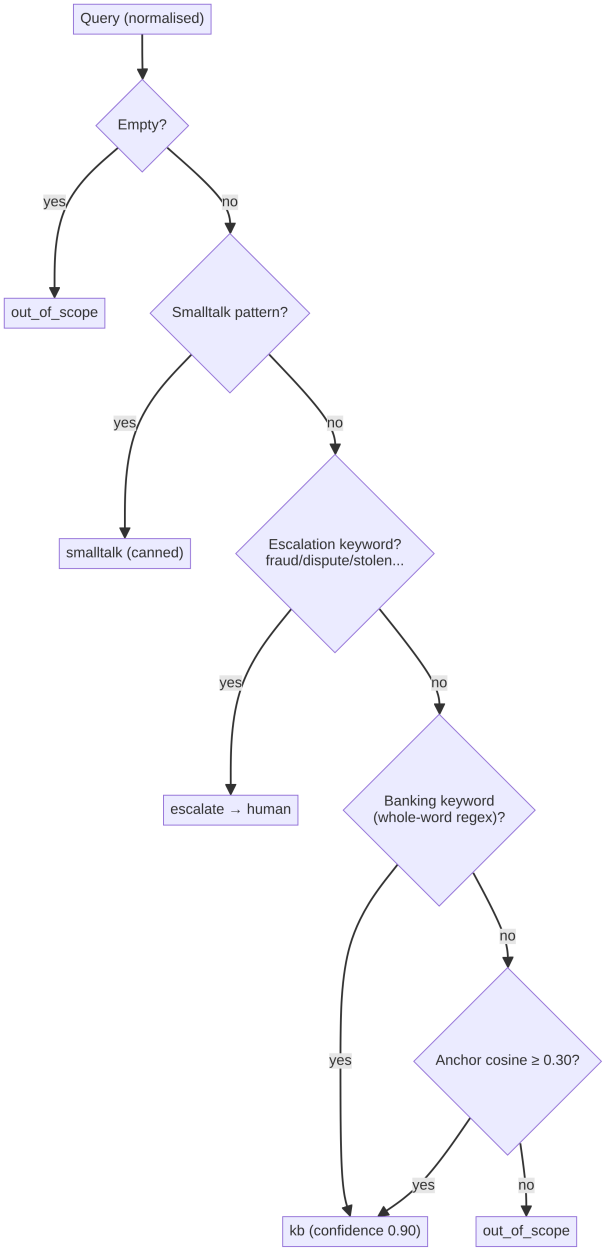
15. API Flow


/api/v1 (JWT required)
POST /chat
Orchestrator → agents
WS /ws/chat
POST /upload
Ingestion → queue → workers
GET /documents
POST /auth/token
access+refresh JWT
DELETE /documents/{id}
POST /reports
Reporting → analytics store
GET /dashboard/metrics
Dashboard → aggregates (cached)
GET /health
POST /feedback
Client

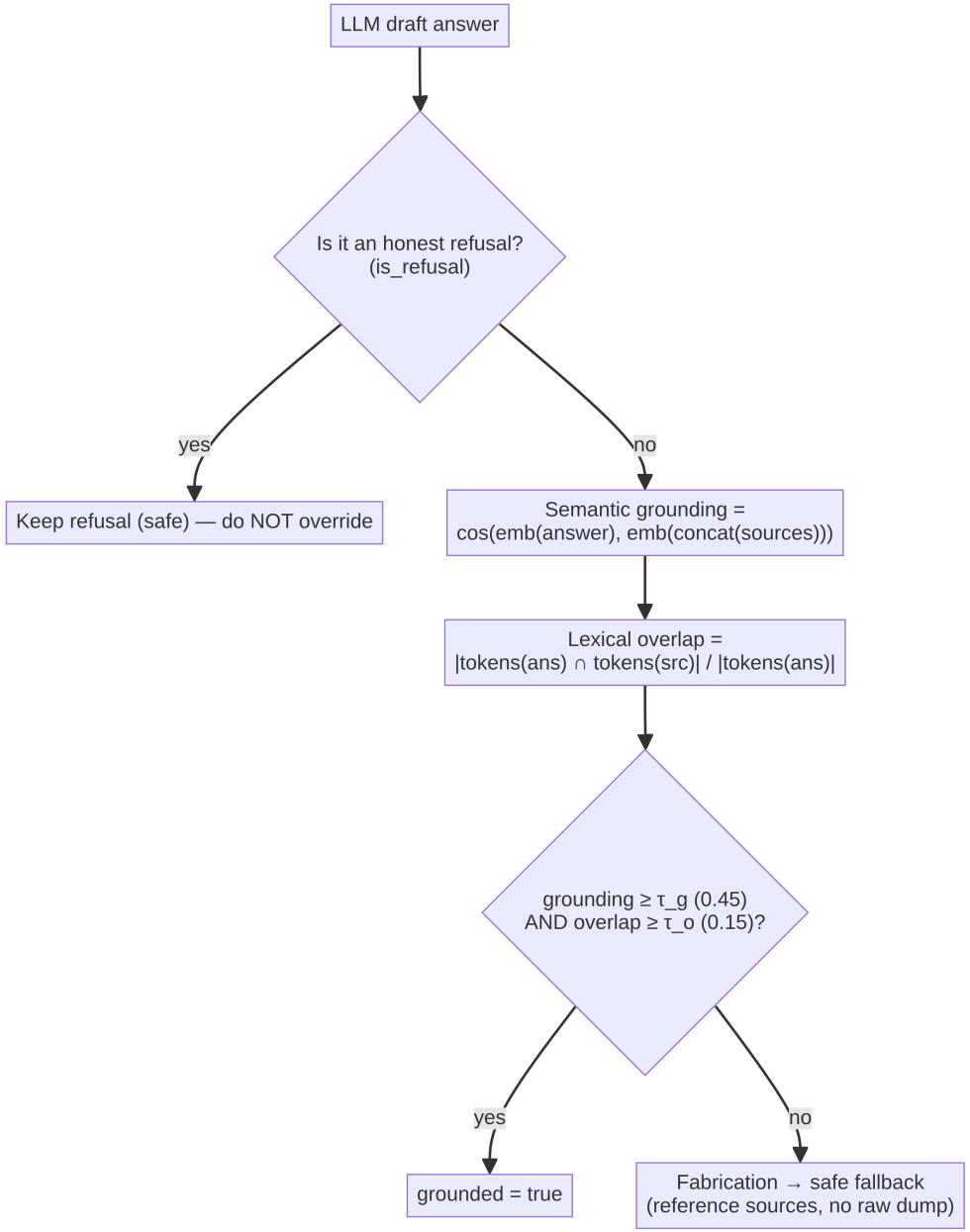
19. Caching


Query
Semantic cache
(embedding cosine ≥ 0.97)
hit
Cached grounded answer
miss
Retrieval cache
(query hash)
hit
miss
Embedding cache
hit
miss
Embed + cache
ANN search
Reuse chunks → synthesise
Store in caches (TTL + invalidation)

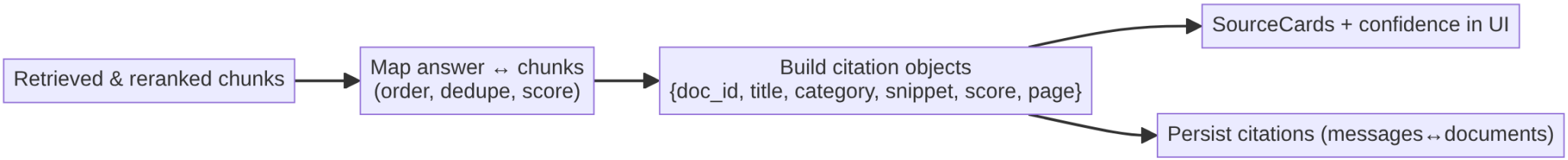
20. Query Routing



21. Hallucination Detection



23. Source Citation Generation



24. Observability

